

Fig. 3: Actual-size PCB of the solar lamp

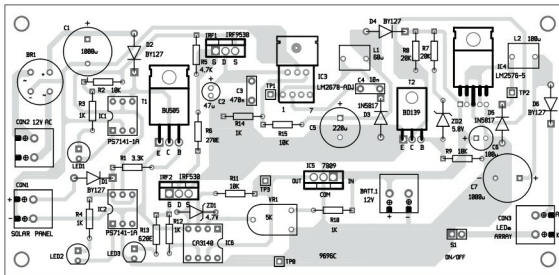


Fig. 4: Component layout of the solar lamp

Test Points

Test point	Details
TP0	0V
TP1	14V
TP2	5V
TP3	9V

Let us see how the lamp works with AC mains with battery voltage lower than 10.5V. 230V AC is stepped down and rectified by a bridge rectifier. DC voltage is around 15V. On account of positive voltage at pin 1 of IC1, the internal LED of IC1 is forward-biased. The relay is switched on. The voltage at pin 5 of IC1 is

transferred to its output pin 6. On state of the red LED indicates that the operation is on AC mains.

Further, this voltage is given to the source terminal of IRF1. As the battery voltage is low, output of comparator circuit built around IC CA4130 is high; when gate of IRF2 is positive with respect to source, it gets switched on, bringing low voltage at gate IRF1 and switching it on. Due to this action, the source of IRF1 gets connected to the drain, transferring the source terminal voltage to the drain terminal. This voltage, in turn, is given to input pin 2 of IC3. By setting the values of resistors at pin 6,

14V is generated at output pin 1 of IC3, which is used for charging the 12V battery. Output of IC3 at pin 1 is input to IC4 at pin 1. Output of IC4 at pin 2 is 5V, which is used to power the lamp load when S1 is on.

Now, let us see how it operates with a solar panel with battery voltage lower than 10.5V.

The action is similar to the operation with AC mains, with a difference that now second solid-state relay IC2 will be operational. On state of the blue LED indicates that the solar panel is in operation.

During day time, solar power is available,

and we would like to make use of this power instead of AC mains. This is achieved by the switching circuit built around transistor T1. The solar panel voltage is more than the DC voltage provided by AC mains circuit. The transistor goes into saturation, voltage at pin 1 of solid-state relay IC1 associated with AC mains circuit is zero, so it becomes non-operational. The AC mains circuit is out of action.

During operation with a 12V DC battery with AC panel, the solar panel is disconnected. Battery voltage is applied to input pin 1 of IC4, which provides 5V to power the LED array.